THE 1600 PROJECT · FOUNDATIONS

F1
Structure and Shortcuts

WEEK	LEVEL	SESSION
1 of 4	Post–Algebra 1 · SAT easy to medium	60 minutes

AT A GLANCE

Goals for this lesson

Students will be able to:

- Solve for an expression like 2x + 3 without solving for x first.
- Use the GCF, difference of squares and perfect squares to simplify or solve.
- Apply exponent rules to variable expressions, including negative exponents.
- Match coefficients when two expressions are equal for all x.

60-minute session

10 min	Warm-up
15 min	Mini-lesson
15 min	Guided practice
15 min	Independent practice
5 min	Exit ticket

MINI-LESSON · 15 MIN

Six patterns worth memorizing

PATTERN	EXAMPLE
Difference of squares: $a^2 - b^2 = (a + b)(a - b)$	$x^2 - 49 = (x + 7)(x - 7)$
Perfect square: $(a + b)^2 = a^2 + 2ab + b^2$	$(x + 5)^2 = x^2 + 10x + 25$
Pull out the greatest common factor	$6x^3 - 9x^2 = 3x^2(2x - 3)$
Solve for the expression, not the variable	$4x + 6 = 18$, so $2x + 3 = 9$
Equal for all x: match the coefficients	$ax + 12 = 3(x + b)$: $a = 3$, $b = 4$
Exponent rules with variables	$(x^3)^2 \cdot x^{-4} = x^6 \cdot x^{-4} = x^2$

MINI-LESSON

Look before you solve

ASK THREE QUESTIONS FIRST

1. Is the quantity they ask for hiding inside the equation? Scale the equation to get it.
2. Does anything factor: a GCF, a difference of squares, a perfect square?
3. Does it say “for all x” or “equivalent”? Expand and match coefficients.

WHY IT MATTERS

See the structure: 30 seconds. Miss it: 2 minutes or more.

Some of these questions can't be solved for x at all; structure is the only way in.

WORKED EXAMPLE 1 · SOLVE FOR THE EXPRESSION

Two variables, one equation

PROBLEM

If 6x − 4y = 18, what is the value of 3x − 2y?

SOLUTION

3x − 2y is exactly half of 6x − 4y.

(6x − 4y) ÷ 2 = 18 ÷ 2

3x − 2y = 9

There's no way to find x and y separately here, so structure is the only route.

WORKED EXAMPLE 2 · DIFFERENCE OF SQUARES

Factor, then divide

PROBLEM

If $x^2 - y^2 = 48$ and $x + y = 8$, what is the value of $x - y$?

SOLUTION

Factor the first equation:

$$(x + y)(x - y) = 48$$

$$8(x - y) = 48$$

$$x - y = 6$$

WORKED EXAMPLE 3 · MATCH COEFFICIENTS

True for all x

PROBLEM

$$(2x + a)(x - 3) = 2x^2 + bx - 15$$

The equation is true for all values of x. What is the value of b?

SOLUTION

Expand the left side:

$$2x^2 + (a - 6)x - 3a$$

Constants: $-3a = -15$, so $a = 5$.

x-terms: $b = a - 6$.

$$\mathbf{b = -1}$$

PRACTICE

Guided, then independent

GUIDED · 15 MIN

Structure hunt

- Show 6 problems for 20 seconds each.
- Before anyone solves, students name the pattern: scale it, factor it, or match coefficients.
- Then solve together and compare times.

INDEPENDENT · 15 MIN

Timed set of 10

3 solve-for-the-expression

3 factoring patterns

2 exponent rules

2 match coefficients

COMMON TRAPS

Where easy points get lost

01 Solving for x when the question asks for an expression.

02 Writing $(a + b)^2$ as $a^2 + b^2$, dropping the $2ab$.

03 Sign errors when expanding before matching coefficients.

04 Reading x^{-2} as a negative number instead of $1/x^2$.

WRAP-UP · 5 MIN

Exit ticket

ON YOUR OWN

1. If $5a + 10b = 35$, what is $a + 2b$?

2. $(x + k)^2 = x^2 + 12x + c$ for all x. What is c?

Source material

- College Board Question Bank: Algebra and Advanced Math, Easy and Medium
- Desmos Toolkit · Advanced Math, for checking coefficient matches

THE 1600 PROJECT · FOUNDATIONS

F2
Lines in Every Form

WEEK	LEVEL	SESSION
2 of 4	Post–Algebra 1 · SAT easy to medium	60 minutes

AT A GLANCE

Goals for this lesson

Students will be able to:

- Find a slope from two points, a table, a graph or any form of the equation.
- Write a line's equation from a point and a slope, or from two points.
- Use parallel (equal) and perpendicular (negative reciprocal) slopes.
- Read and use f(x) values from tables and equations.

60-minute session

10 min	Warm-up
15 min	Mini-lesson
15 min	Guided practice
15 min	Independent practice
5 min	Exit ticket

MINI-LESSON · 15 MIN

Getting the line from anything

YOU'RE GIVEN	WHAT TO DO
Two points	$m = (y_2 - y_1) / (x_2 - x_1)$, then plug in a point to find b
Standard form Ax + By = C	slope = −A/B; y-intercept = C/B
A table	slope = change in y ÷ change in x between any two rows
A point and a slope	$y - y_1 = m(x - x_1)$, then simplify
Intercepts	x-intercept: set y = 0; y-intercept: set x = 0

MINI-LESSON

Parallel, perpendicular and f(x)

Parallel

Same slope, different intercept.

$y = 2x + 1$ and $y = 2x - 7$

Perpendicular

Negative reciprocal slopes; they multiply to −1.

$2/3$ and $-3/2$

Function notation

$f(a) = b$ means the point (a, b) is on the graph.

input a, output b

SPEED HABIT Get the slope first, in whatever form is fastest. Most line questions are decided by the slope alone.

WORKED EXAMPLE 1 · TWO POINTS

Find the y-intercept

PROBLEM

A line passes through the points (2, 5) and (6, 17). What is the y-intercept of the line?

SOLUTION

$$m = (17 - 5) / (6 - 2) = 3$$

Plug in (2, 5):

$$5 = 3(2) + b$$

$$b = -1, \text{ so } (0, -1)$$

WORKED EXAMPLE 2 · PERPENDICULAR

Flip it and switch the sign

PROBLEM

Line k is perpendicular to $2x - 3y = 12$ and passes through $(4, 1)$. What is the equation of line k?

SOLUTION

Given slope: $-A/B = -2/(-3) = 2/3$.

Perpendicular slope: $-3/2$.

$$y - 1 = -(3/2)(x - 4)$$

$$y = -(3/2)x + 7$$

Check: $-(3/2)(4) + 7 = 1$.

WORKED EXAMPLE 3 · TABLE AND f(x)

Input or output?

PROBLEM

The linear function f has these values:

x	0	2	4
f(x)	7	13	19

(a) What is f(10)? (b) If f(a) = 40, what is a?

SOLUTION

Slope = 6 ÷ 2 = 3; f(0) = 7.

f(x) = 3x + 7

(a) f(10) = 37

3a + 7 = 40, so 3a = 33.

(b) a = 11

PRACTICE

Guided, then independent

GUIDED · 15 MIN

Four-way match

- Cards show 5 lines, each as a graph, a table, an equation and a context.
- Groups match all four versions of each line, using slope and intercept as evidence.

INDEPENDENT · 15 MIN

Timed set of 10

3 slope and intercepts

3 write the equation

2 parallel or perpendicular

2 table or f(x) notation

COMMON TRAPS

Where easy points get lost

01 Putting x on top: slope is change in y over change in x.

02 Dropping the negative in −A/B for standard form.

03 Perpendicular: flipping without switching the sign, or the reverse.

04 Mixing up f(a), an output, with a, an input.

WRAP-UP · 5 MIN

Exit ticket

ON YOUR OWN

1. Write the line through (−1, 4) that is parallel to y = −2x + 9.

2. What slope is perpendicular to 4x + y = 3?

Source material

- College Board Question Bank: Algebra, Easy and Medium
- Desmos Toolkit · Algebra, for regression from points and tables

THE 1600 PROJECT · FOUNDATIONS

F3

Word Problems to Equations

WEEK	LEVEL	SESSION
3 of 4	Post–Algebra 1 · SAT easy to medium	60 minutes

AT A GLANCE

Goals for this lesson

Students will be able to:

- Translate key phrases into expressions, equations and inequalities.
- Build and solve a system of two equations from a word problem.
- Write and interpret inequalities, rounding to fit the context.
- Pick the equation or function that matches a situation.

60-minute session

10 min	Warm-up
15 min	Mini-lesson
15 min	Guided practice
15 min	Independent practice
5 min	Exit ticket

MINI-LESSON

Words into symbols

THE WORDS SAY	THE MATH IS	EXAMPLE
is, equals, results in	**=**	Total is 40 → … = 40
more than, increased by	**+**	7 more than n → n + 7
less than, fewer than	**− (order flips)**	5 less than x → x − 5
of, times, product	**×**	30% of p → 0.3p
per, for each, every	**rate × amount**	$4 per hour → 4h
at least / at most	**≥ / ≤**	at least 20 → ≥ 20
twice as many A as B	**A = 2B**	twice as many cats as dogs → c = 2d

MINI-LESSON

Define, translate, solve, check

1

Define

Name each variable in words: b = number of boxes.

2

Translate

One sentence at a time, using the phrase table.

3

Solve

Algebra, or Desmos for systems and inequalities.

4

Sanity-check

Does the answer fit the story? Whole number? Right variable?

MODEL QUESTIONS

Many questions only ask which equation fits. Match the starting amount to the constant and the rate to the coefficient. Don't solve.

WORKED EXAMPLE 1 · INEQUALITY

Round the way the story needs

PROBLEM

An elevator can carry at most 1,200 pounds. The operator weighs 180 pounds and each box weighs 45 pounds. What is the greatest number of boxes the operator can take in one trip?

SOLUTION

At most → ≤

180 + 45b ≤ 1200

45b ≤ 1020

b ≤ 22.67, and b must be a whole number.

22 boxes

WORKED EXAMPLE 2 · SYSTEM

Two facts, two equations

PROBLEM

A student buys 12 school supplies. Notebooks cost $3 each and binders cost $7 each. She spends $56 in total. How many binders did she buy?

SOLUTION

n + b = 12

3n + 7b = 56

Substitute n = 12 − b: 36 − 3b + 7b = 56, so 4b = 20.

b = 5 binders

Check: 7 notebooks, 3(7) + 7(5) = 56.

WORKED EXAMPLE 3 · MODELING

Which expression fits?

PROBLEM

A phone plan costs $30 per month and includes 200 minutes. Each minute over 200 costs $0.10. Which expression gives the monthly cost, in dollars, for t minutes, where t > 200?

A) 30 + 0.10t
B) 30 + 0.10(t − 200)
C) 0.10(t − 200)
D) 30t + 0.10

SOLUTION

Only the minutes over 200 are charged extra.

Extra minutes: t − 200

Cost = 30 + 0.10(t − 200)

B

Test t = 300: 30 + 0.10(100) = $40.

PRACTICE

Guided, then independent

GUIDED · 15 MIN

Setup only

- Six stories; students write the equation or inequality but don't solve.
- Partners swap and check by plugging a test number into each other's setup.
- Discuss any setup two groups wrote differently.

INDEPENDENT · 15 MIN

Timed set of 10

3 inequalities in context

3 systems from word problems

4 which model fits

COMMON TRAPS

Where easy points get lost

01 Writing 5 − x for “5 less than x”. The order flips: x − 5.

02 Rounding the wrong way. “At most” and “greatest” mean round down; “at least” means round up.

03 Reversing “twice as many A as B”. It is A = 2B; test with small numbers.

04 Solving for the wrong variable. Circle what the question actually asks for.

WRAP-UP · 5 MIN

Exit ticket

ON YOUR OWN

1. A gym charges a $50 sign-up fee plus $20 per month. Kai has $250. What is the greatest number of months he can pay for?

2. A field trip has 3 times as many students as teachers, with 52 people in total. How many teachers are going?

Source material

- College Board Question Bank: Algebra and Problem-Solving and Data Analysis, Easy and Medium
- Desmos Toolkit · Algebra, for checking systems and inequalities

THE 1600 PROJECT · FOUNDATIONS

F4

Geometry and Data Quick Wins

WEEK	LEVEL	SESSION
4 of 4	Post–Algebra 1 · SAT easy to medium	60 minutes

AT A GLANCE

Goals for this lesson

Students will be able to:

- Use angle facts for lines, triangles and parallel lines.
- Apply 45-45-90, 30-60-90 and Pythagorean triples without the full theorem.
- Find circle area and circumference from either r or d.
- Reason about mean, median and two-way tables using sums and the right denominator.

60-minute session

10 min	Warm-up
15 min	Mini-lesson
15 min	Guided practice
15 min	Independent practice
5 min	Exit ticket

MINI-LESSON · GEOMETRY

Six facts worth memorizing

Angles

Straight line = 180°
Triangle = 180°

Vertical angles are equal.

Parallel lines

Corresponding and alternate angles are equal

Same-side interior angles add to 180°.

45-45-90

x, x, x√2

Isosceles right triangle; half a square.

30-60-90

x, x√3, 2x

Shortest side is opposite 30°; hypotenuse is double it.

Pythagorean triples

3-4-5 · 5-12-13 · 8-15-17

Multiples work too: 6-8-10, 9-12-15.

Circles

A = πr² · C = 2πr

Diameter = 2r. Find r first.

MINI-LESSON · DATA

Statistics quick wins

Mean vs median

- An outlier pulls the mean toward it; the median barely moves.
- Mean × count = sum. Work with sums when values are added or removed.
- Skewed right: mean > median.

Two-way tables

	Yes	No	Total
9th	18	12	**30**
10th	14	26	40

P(yes **given** 9th) = 18/30. The word “given” picks the denominator.

WORKED EXAMPLE 1 · SPECIAL RIGHT TRIANGLE

No Pythagorean theorem needed

PROBLEM

In right triangle ABC, angle A measures 30° and the hypotenuse is 10. What are the lengths of the two legs?

SOLUTION

30-60-90 sides: x, x√3, 2x

2x = 10, so x = 5

Short leg is opposite 30°; long leg is opposite 60°.

Legs: 5 and 5√3

WORKED EXAMPLE 2 · PARALLEL LINES

Same-side angles add to 180°

PROBLEM

Two parallel lines are cut by a transversal. A pair of same-side interior angles measure (3x + 10)° and (2x + 20)°. What are the two angle measures?

SOLUTION

(3x + 10) + (2x + 20) = 180

5x + 30 = 180, so x = 30

3(30) + 10 and 2(30) + 20

100° and 80°

WORKED EXAMPLE 3 · MEAN

Think in sums

PROBLEM

The mean of five numbers is 12. One of the numbers, 20, is removed. What is the mean of the remaining four numbers?

SOLUTION

Mean × count = sum

Sum = 12 × 5 = 60

New sum = 60 − 20 = 40

40 ÷ 4 = 10

New mean: 10

PRACTICE

Guided, then independent

GUIDED · 15 MIN

Reference sheet hunt

- Students find each of today's six facts on the SAT reference sheet (or notice which ones aren't there).
- Six quick problems: partners race to answer, then explain which fact they used.

INDEPENDENT · 15 MIN

Mixed review of 12

3 F1 · structure and shortcuts

3 F2 · lines in every form

3 F3 · word problems

3 F4 · geometry and data

COMMON TRAPS

Where easy points get lost

01 Trusting the diagram. SAT figures aren't always drawn to scale; use the given numbers.

02 Mixing up 30-60-90 sides. The side opposite 30° is the shortest; the hypotenuse is twice it.

03 Using the diameter as the radius. Halve it first, then square.

04 Dividing by the old count after removing or adding a data value.

WRAP-UP · 5 MIN

Exit ticket

ON YOUR OWN

1. An isosceles right triangle has legs of length 6. What is the length of the hypotenuse?

2. A circle has circumference 10π. What is its area?

Source material

- College Board Question Bank: Geometry and Trigonometry and Problem-Solving and Data Analysis, Easy and Medium
- SAT math reference sheet (in Bluebook)

THE 1600 PROJECT · CATEGORY 1: ALGEBRA

1.1 Linear Equations and Systems

WEEKS	SAT DOMAIN	SESSION
3 to 4	Algebra · about 35% of Math	60 minutes

AT A GLANCE

Goals for this lesson

Students will be able to:

- Translate a word problem into a linear equation and interpret the slope and intercept in context.
- Solve a system of two linear equations by substitution, elimination, or Desmos.
- Determine when a system has one solution, no solution, or infinitely many.
- Solve and graph linear inequalities, flipping the sign when multiplying or dividing by a negative.

60-minute session

10 min	Warm-up
15 min	Mini-lesson
15 min	Guided practice
15 min	Independent practice
5 min	Exit ticket

MINI-LESSON · 15 MIN

Three forms of a line

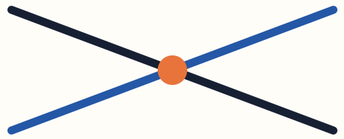
FORM	EQUATION	WHAT YOU READ OFF IT
Slope-intercept	$y = mx + b$	slope m, y-intercept b
Standard	$Ax + By = C$	slope = −A/B
Slope from two points	$m = (y_2 - y_1) / (x_2 - x_1)$	rate of change

IN WORD PROBLEMS

Slope is the **rate** (“per month,” “each”). The intercept is the **starting value** (“fee,” “initial”).

MINI-LESSON

How many solutions?



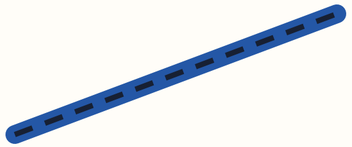
One solution

Different slopes: the lines cross once.



No solution

Same slope, different intercepts: parallel lines.



Infinitely many

One equation is a multiple of the other: same line.

TRANSLATION METHOD: NAME, RATE, START

1 Name the variable → 2 Rate = the coefficient → 3 Start = the constant → 4 Write the equation

WORKED EXAMPLE 1 · TRANSLATION

Gym membership

PROBLEM

A gym charges a $40 sign-up fee plus $25 per month. Jordan paid $240 in total. For how many months did Jordan pay?

SOLUTION

Rate = 25 per month. Start = 40.

$$40 + 25m = 240$$

$$25m = 200$$

$$m = 8 \text{ months}$$

WORKED EXAMPLE 2 · SYSTEM

Play tickets

PROBLEM

A school sold 150 tickets to a play. Adult tickets cost $12 and student tickets cost $8. Ticket sales totaled $1,480. How many adult tickets were sold?

SOLUTION

$a + s = 150$ and $12a + 8s = 1{,}480$

Substitute $s = 150 - a$:

$$12a + 8(150 - a) = 1{,}480$$

$$4a + 1{,}200 = 1{,}480$$

$a = 70$ adult tickets ($s = 80$)

Desmos check: graph $x + y = 150$ and $12x + 8y = 1480$, then click the intersection (70, 80).

WORKED EXAMPLE 3 · NUMBER OF SOLUTIONS

Find k for no solution

PROBLEM

For what value of k does the system below have no solution?

$$3x + ky = 7$$

$$6x + 10y = 5$$

SOLUTION

No solution means parallel lines. Multiply the first equation by 2:

$$6x + 2ky = 14$$

The x-coefficients match, so the y-coefficients must match too:

$$2k = 10, \text{ so } k = 5$$

The constants (14 and 5) differ, so the lines are parallel, not identical.

PRACTICE

Guided, then independent

GUIDED · 15 MIN

6 word problems as a group

- Before writing anything, students say the rate and the starting value.
- Solve at least two systems both by hand and in Desmos.

INDEPENDENT · 15 MIN

Timed set of 10

About 95 seconds per question, SAT pace.

4	4	2
translation	systems	inequalities

COMMON TRAPS

Where points get lost

01 Answering the wrong variable: solving for s when the question asks for a.

02 Swapping slope and intercept in context questions.

03 Forgetting to flip the inequality when dividing by a negative.

04 Confusing “no solution” and “infinitely many.”

WRAP-UP · 5 MIN

Exit ticket

ON YOUR OWN

Solve one context problem: write the equation, then explain in words what the slope means.

Source material

- College Board Question Bank: Algebra (linear equations in one and two variables; systems of two linear equations)
- Khan Academy SAT Math
- Desmos for graphing systems

2.1 Quadratics and Functions

WEEKS	SAT DOMAIN	SESSION
7 to 9	Advanced Math · about 35% of Math	60 minutes

AT A GLANCE

Goals for this lesson

Students will be able to:

- Solve quadratics by factoring, the quadratic formula, or Desmos, and choose the fastest method.
- Use the discriminant to find how many real solutions a quadratic has.
- Read the vertex, intercepts, and direction of a parabola from its equation, and go from graph to equation.
- Evaluate and transform functions using f(x) notation.

60-minute session

10 min	Warm-up
15 min	Mini-lesson
15 min	Guided practice
15 min	Independent practice
5 min	Exit ticket

MINI-LESSON · 15 MIN

Three forms of a quadratic

FORM	EQUATION	SHOWS YOU
Standard	$y = ax^2 + bx + c$	y-intercept c; vertex at $x = -b/(2a)$
Factored	$y = a(x - r)(x - s)$	x-intercepts r and s
Vertex	$y = a(x - h)^2 + k$	vertex (h, k); min or max value k

$a > 0$

Opens up, so the vertex is a **minimum**.

$a < 0$

Opens down, so the vertex is a **maximum**.

MINI-LESSON · KEEP ON THE BOARD ALL THREE WEEKS

Quadratic formula and discriminant

$$x = (-b \pm \sqrt{(b^2 - 4ac)}) / 2a$$

$b^2 - 4ac > 0$

Two real solutions

$b^2 - 4ac = 0$

Exactly one real solution

$b^2 - 4ac < 0$

No real solutions

SHORTCUTS Sum of the solutions = **$-b/a$** Product of the solutions = **c/a**

MINI-LESSON

Transformations of f(x)

CHANGE	WHAT HAPPENS	EXAMPLE
f(x) + k	Moves up k	f(x) + 3 moves up 3
f(x – h)	Moves right h (opposite of the sign)	f(x + 2) moves left 2
–f(x)	Reflects over the x-axis	$-x^2$ opens down

NOTATION **f(2)** means evaluate: plug in 2. **f(x) = 2** means solve: find the x that gives 2.

WORKED EXAMPLE 1 · FACTORING

Solve by factoring

PROBLEM

Solve:

$$x^2 - 5x - 14 = 0$$

SOLUTION

Find two numbers that multiply to −14 and add to −5: **−7 and 2**.

$$(x - 7)(x + 2) = 0$$

$$x = 7 \text{ or } x = -2$$

Check with the shortcut: the sum is 7 + (−2) = 5, and −b/a = −(−5)/1 = 5.

WORKED EXAMPLE 2 · VERTEX AND TRANSFORMATION

Where does the vertex move?

PROBLEM

$$f(x) = (x - 3)^2 + 4$$

If $g(x) = f(x + 2) - 1$, what is the vertex of g?

SOLUTION

The vertex of f is **(3, 4)**.

f(x + 2) moves it **left 2**: (1, 4).

The − 1 moves it **down 1**: (1, 3).

Vertex of g = (1, 3)

WORKED EXAMPLE 3 · DISCRIMINANT

Exactly one real solution

PROBLEM

For what value of c does this equation have exactly one real solution?

$$x^2 + 6x + c = 0$$

SOLUTION

Exactly one solution means the discriminant is zero:

$$6^2 - 4(1)(c) = 0$$

$$36 = 4c$$

$$c = 9$$

Desmos check: graph $y = x^2 + 6x + c$ with a slider for c and watch the parabola touch the x-axis at c = 9.

PRACTICE

Guided, then independent

GUIDED · 15 MIN

Graph-to-equation matching cards

- In pairs, match 8 quadratic equations to 8 graphs.
- Justify each match using the vertex, the intercepts, or the direction it opens.

INDEPENDENT · 15 MIN

Timed set of 10

3	factoring or quadratic formula
3	vertex or graph
2	discriminant
2	function notation or transformation

COMMON TRAPS

Where points get lost

01 Sign errors in vertex form: $(x - 3)^2$ means $h = 3$, not -3.

02 Transformation direction: $f(x + 2)$ moves left, not right.

03 Dropping the ± in the quadratic formula.

04 $f(2)$ versus $f(x) = 2$: evaluating versus solving.

WRAP-UP · 5 MIN

Exit ticket

ON YOUR OWN

For one quadratic in standard form, give its vertex, its number of real solutions, and its y-intercept.

Source material

- College Board Question Bank: Advanced Math (equivalent expressions; nonlinear equations and systems; nonlinear functions)
- Khan Academy walkthroughs on factoring, as pre-work
- Desmos to check roots and vertices

2.2

Exponents and Exponential Models

WEEKS
10 to 11

SAT DOMAIN
Advanced Math · about 35% of Math

SESSION
60 minutes

AT A GLANCE

Goals for this lesson

Students will be able to:

- Simplify expressions using exponent rules, including negative and fractional exponents.
- Write and interpret growth and decay models of the form $y = a(1 \pm r)^t$.
- Tell linear from exponential relationships in tables and context.
- Convert between radicals and fractional exponents.

60-minute session

10 min	Warm-up
15 min	Mini-lesson
15 min	Guided practice
15 min	Independent practice
5 min	Exit ticket

MINI-LESSON · 15 MIN

Exponent rules to post

RULE	EXAMPLE
$x^a \cdot x^b = x^{a+b}$	$x^3 \cdot x^4 = x^7$
$x^a / x^b = x^{a-b}$	$x^6 / x^2 = x^4$
$(x^a)^b = x^{ab}$	$(x^3)^2 = x^6$
$x^{-a} = 1 / x^a$	$x^{-2} = 1/x^2$
$x^{(m/n)} = \sqrt[n]{(x^m)}$	$8^{(2/3)} = 4$
$x^0 = 1$	$5^0 = 1$

RADICALS

A fractional exponent is a root. The bottom number is the root; the top number is the power.

$8^{(2/3)} = (\sqrt[3]{8})^2 = 2^2 = 4$

MINI-LESSON

The exponential model

$$y = a(1 \pm r)^t$$

a = starting amount · **r** = percent rate as a decimal (+ growth, − decay) · **t** = time

6% increase

× 1.06

15% decrease

× 0.85

Doubles every d years

y = a · 2^(t/d)

LINEAR

Adds the same amount each step (constant difference).

EXPONENTIAL

Multiplies by the same factor each step (constant ratio).

WORKED EXAMPLE 1 · RULES

Simplify the expression

PROBLEM

Simplify:

$$(2x^3)^2 \cdot x^4 / (4x^5)$$

SOLUTION

$$(2x^3)^2 = 4x^6$$

$$4x^6 \cdot x^4 = 4x^{10}$$

$$4x^{10} / (4x^5) = x^5$$

Answer: x^5

Note: the 2 is squared too. $(2x^3)^2$ is $4x^6$, not $2x^6$.

WORKED EXAMPLE 2 · DECAY

Car depreciation

PROBLEM

A car is bought for $24,000 and loses 15% of its value each year. What is it worth after 2 years?

SOLUTION

A 15% loss keeps 85%, so the factor is **0.85**.

$$V = 24{,}000(0.85)^2$$

$$= 24{,}000 \times 0.7225$$

$$= \$17{,}340$$

WORKED EXAMPLE 3 · DOUBLING

Bacteria culture

PROBLEM

A bacteria culture starts with 500 cells and doubles every 3 hours. How many cells are there after 12 hours?

SOLUTION

Doubling every 3 hours means the exponent is **t/3**.

$$N = 500 \cdot 2^{(12/3)}$$

$$= 500 \cdot 2^4$$

$$= 8{,}000 \text{ cells}$$

PRACTICE

Guided, then independent

GUIDED · 15 MIN

Linear or exponential?

- Sort 8 context cards and tables into “linear” and “exponential,” then write the model for each.
- Work 4 growth or decay problems together, checking each in Desmos with a table.

INDEPENDENT · 15 MIN

Timed set of 10

4 exponent and radical simplifications

4 real-world growth or decay problems

2 linear-versus-exponential items

COMMON TRAPS

Where points get lost

01 Using 0.15 instead of 0.85 for a 15% decrease.

02 Adding exponents when you should multiply (power of a power).

03 Misreading the time unit: “doubles every 3 hours” means t/3, not 3t.

04 Treating percent growth as linear (adding 5% of the original each year).

WRAP-UP · 5 MIN

Exit ticket

ON YOUR OWN

Write the model for “a town of 8,000 people grows 3% per year,” and state what 1.03 means in context.

Source material

- College Board Question Bank: Advanced Math (equivalent expressions; nonlinear equations and systems; nonlinear functions)
- Khan Academy walkthroughs on exponential models, as pre-work
- Desmos tables to check growth and decay

3.1 Ratios, Rates, and Percentages

WEEKS	SAT DOMAIN	SESSION
5 to 6	Data Analysis · about 15% of Math	60 minutes

AT A GLANCE

Goals for this lesson

Students will be able to:

- Set up and solve proportions from word problems.
- Convert units by multiplying by conversion fractions so units cancel.
- Compute percent change and handle successive percent changes with multipliers.

60-minute session

10 min	Warm-up
15 min	Mini-lesson
15 min	Guided practice
15 min	Independent practice
5 min	Exit ticket

MINI-LESSON · 15 MIN

Three tools for the board

1

Proportion

Keep units in the same place on both sides.

part / whole = part / whole

2

Unit conversion

“Railroad tracks”: write each conversion as a fraction equal to 1, and cancel units until only the target units remain.

3

Percent as a multiplier

Increase p%: × (1 + p/100)
Decrease p%: × (1 − p/100)
Successive changes multiply.

PERCENT CHANGE (new − old) / old × 100

WORKED EXAMPLE 1 · PROPORTION

Scaling a recipe

PROBLEM

A recipe uses 3 cups of flour to make 24 cookies. How much flour is needed for 40 cookies?

SOLUTION

Cups on top, cookies on the bottom, on both sides:

$$\frac{3}{24} = \frac{x}{40}$$

$$24x = 120$$

$$x = 5 \text{ cups}$$

WORKED EXAMPLE 2 · UNIT CONVERSION

Miles per hour to feet per second

PROBLEM

A car travels at 54 miles per hour. What is its speed in feet per second?

(1 mile = 5,280 feet)

SOLUTION · RAILROAD TRACKS

$$\frac{54\ \text{mi}}{1\ \text{hr}} \times \frac{5{,}280\ \text{ft}}{1\ \text{mi}} \times \frac{1\ \text{hr}}{3{,}600\ \text{s}}$$

Miles cancel, hours cancel, and feet per second remain.

= 79.2 feet per second

WORKED EXAMPLE 3 · SUCCESSIVE PERCENTS

Up 20%, then down 20%

PROBLEM

A jacket's price is raised 20%, then the new price is lowered 20%. What is the overall change?

SOLUTION

Successive changes multiply:

1.20 × 0.80 = 0.96

The final price is 96% of the original.

A 4% decrease, not zero

PRACTICE

Guided, then independent

GUIDED · 15 MIN

6 applied word problems as a group

- Students must write units at every step of each conversion.

INDEPENDENT · 15 MIN

Timed set of 10

3 proportions

3 unit conversions

4 percent problems, including one “percent of a percent” and one reverse problem

Reverse example: after a 25% discount the price is $60. What was the original?

COMMON TRAPS

Where points get lost

01 Dividing by the new value instead of the original in percent change.

02 Adding successive percents: +20% then −20% is not 0%.

03 Flipping a conversion fraction so the units don't cancel.

04 Reverse percents: $60 after 25% off means 0.75x = 60, so x = 80, not 60 × 1.25.

WRAP-UP · 5 MIN

Exit ticket

ON YOUR OWN

Solve one reverse-percent problem and one unit conversion, with units shown at every step.

Source material

- College Board Question Bank: Problem-Solving and Data Analysis
- Khan Academy for fundamentals
- Real public datasets (school data, census tables) for extra practice, as long as questions mirror SAT phrasing

3.2
Statistics and Probability

WEEKS	SAT DOMAIN	SESSION
12 to 14	Data Analysis · about 15% of Math	60 minutes

AT A GLANCE

Goals for this lesson

Students will be able to:

- Compute mean and median and predict how an outlier or added value changes each.
- Compare the spread (standard deviation) of two data sets without calculating it.
- Find simple and conditional probabilities from a two-way table.
- Judge what a study's sampling and design allow you to conclude.

60-minute session

10 min	Warm-up
15 min	Mini-lesson
15 min	Guided practice
15 min	Independent practice
5 min	Exit ticket

MINI-LESSON · 15 MIN

Five key ideas to post

Center

The mean is pulled toward outliers. The median is resistant to them.

Spread

Standard deviation measures spread around the mean. More values far from the center means a larger SD. The SAT asks you to compare, not compute.

Probability

Favorable outcomes ÷ total outcomes. For “given that…”, the denominator is only the given group.

Sampling

Results from a random sample generalize to the population it was drawn from, within the margin of error.

Causation

Only a **randomly assigned experiment** supports a cause-and-effect claim.

WORKED EXAMPLE 1 · OUTLIERS

Removing an outlier

PROBLEM

Data set:

4, 6, 7, 7, 9, 31

What happens to the mean and median if 31 is removed?

SOLUTION

	MEAN	MEDIAN
With 31	64/6 ≈ 10.7	7
Without 31	33/5 = 6.6	7

The mean drops by about 4.1. The median doesn't change.

WORKED EXAMPLE 2 · CONDITIONAL PROBABILITY

A survey of 200 students

PROBLEM

	PLAYS A SPORT	NO SPORT	TOTAL
Grade 11	45	55	100
Grade 12	**60**	40	100
Total	**105**	95	200

If a student who plays a sport is chosen at random, what is the probability they are in grade 12?

SOLUTION

“Given that they play a sport” means only the sport column counts.

Denominator: **105**, not 200.

P = 60 / 105

= 4/7

WORKED EXAMPLE 3 · INFERENCE

What can the poll tell us?

PROBLEM

A random sample of 400 voters in a city found that 58% support a new park, with a margin of error of 4 percentage points. Which conclusion is best supported?

SOLUTION

Supported

True support among city voters is plausibly between 54% and 62%.

Not supported

- Anything about voters in other cities or the state
- That exactly 58% of all city voters support it

PRACTICE

Guided, then independent

GUIDED · 15 MIN

2 two-way tables, 2 study designs

For each scenario, students answer two questions:

1. **Was it random sampling?** Who can we generalize to?
2. **Was it random assignment?** Can we claim cause?

INDEPENDENT · 15 MIN

Timed set of 10

3 measures of center

2 standard-deviation comparisons

3 probability

2 inference

COMMON TRAPS

Where points get lost

01 Using the wrong denominator in conditional probability.

02 Generalizing beyond the sampled population.

03 Claiming causation from an observational study.

04 Assuming the mean equals the median in skewed data.

WRAP-UP · 5 MIN

Exit ticket

ON YOUR OWN

Answer one conditional-probability question from a table, and write one sentence on what a given study can and can't conclude.

Source material

- College Board Question Bank: Problem-Solving and Data Analysis
- Khan Academy for statistics fundamentals
- Real public datasets (school data, census tables) for extra practice, as long as questions mirror SAT phrasing

3.3
Data and Graph Interpretation

WEEKS
21 to 22 (capstone)

SAT DOMAIN
Data Analysis · about 15% of Math

SESSION
60 minutes

AT A GLANCE

Goals for this lesson

Students will be able to:

- Read titles, axes, units, and scales before reading any values.
- Interpret the slope and y-intercept of a line of best fit in context and use it to predict.
- Compute and interpret a residual (actual minus predicted).
- Tell association from causation in graphed data.

60-minute session

10 min	Warm-up
15 min	Mini-lesson
15 min	Guided practice
15 min	Independent practice
5 min	Exit ticket

MINI-LESSON · 15 MIN

The TASK strategy

Most graph questions are lost by misreading, not by math. Slow down for the first 10 seconds.

T

Title

What is being measured?

A

Axes

What does each axis show, and in what units?

S

Scale

Does each gridline mean 1, 5, or 1,000? Does the axis start at zero?

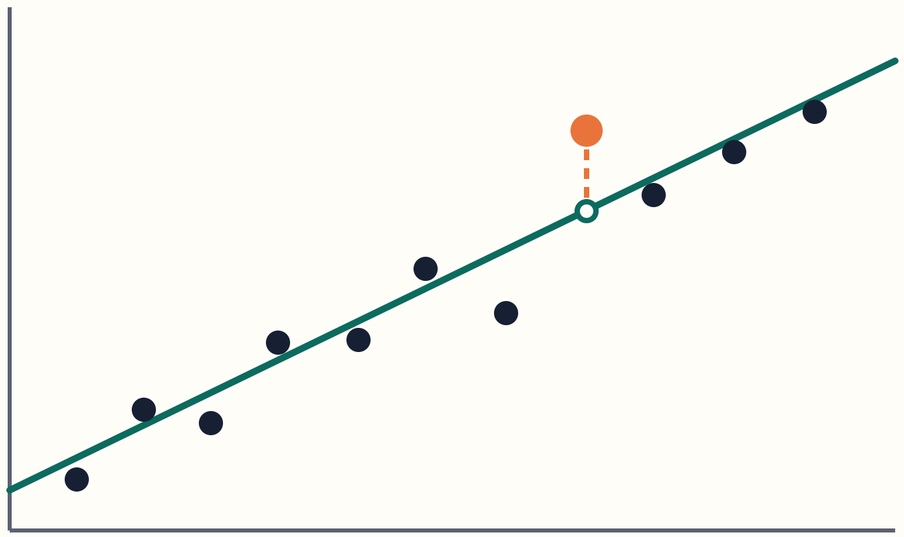
K

Key

What does each color, bar, or symbol stand for?

MINI-LESSON

Reading a line of best fit



Orange point sits above the line, so its residual is positive.

Slope m

The predicted change in y for each one-unit increase in x.

Intercept b

The predicted y when x = 0.

Residual

Actual y – predicted y. Positive above the line, negative below.

WORKED EXAMPLE 1 · LINE OF BEST FIT

Study hours and test scores

PROBLEM

A scatterplot of students' study hours (x) and test scores (y) has the line of best fit:

$$y = 2.5x + 30$$

A student who studied 8 hours scored 55.

SOLUTION

SLOPE

Each additional hour of study is associated with a 2.5-point higher predicted score.

PREDICTION FOR 8 HOURS

$$2.5(8) + 30 = 50$$

RESIDUAL

$$55 - 50 = 5$$

Positive, so the point is above the line.

WORKED EXAMPLE 2 · ASSOCIATION VERSUS CAUSATION

Does studying cause higher scores?

PROBLEM

The same scatterplot shows a strong positive association. Can we conclude that studying more causes higher scores?

SOLUTION

No.

The data is observational. Nobody was randomly assigned to study more.

SUPPORTED

“Students who studied more **tended to** score higher.”

NOT SUPPORTED

Any cause-and-effect claim.

PRACTICE

Guided, then independent

GUIDED · 15 MIN

4 projected graphics

A bar chart, a line graph, a scatterplot, and a table.

- Students do TASK aloud before any question is read.
- Then answer 2 questions per graphic.

INDEPENDENT · 15 MIN

Graphic-heavy timed set

8–10 questions

- Students circle the scale on every graph before answering.

COMMON TRAPS

Where points get lost

01 Misreading the scale, or an axis that doesn't start at zero.

02 Residual sign: predicted minus actual instead of actual minus predicted.

03 Interpreting the y-intercept when x = 0 makes no sense in context.

04 Causal language in answer choices for observational data.

WRAP-UP · 5 MIN

Exit ticket

ON YOUR OWN

Given one scatterplot and its line of best fit, interpret the slope in a sentence and compute one residual.

Source material

- College Board Question Bank: Problem-Solving and Data Analysis
- Khan Academy for statistics fundamentals
- Real public datasets (school data, census tables), as long as questions mirror SAT phrasing

4.1
Geometry and Trigonometry

WEEKS
15 to 17

SAT DOMAIN
Geometry and Trig · about 15% of Math

SESSION
60 minutes

AT A GLANCE

Goals for this lesson

Students will be able to:

- Use the official SAT reference sheet quickly and know what it does not include.
- Find areas and volumes, and use similar triangles (lengths scale by k, areas by k^2, volumes by k^3).
- Apply SOH-CAH-TOA and the fact that $\sin x° = \cos(90 - x)°$.
- Find the center and radius of a circle from its equation, including by completing the square.
- Compute arc length and sector area, and convert between degrees and radians.

60-minute session

10 min	Warm-up
15 min	Mini-lesson
15 min	Guided practice
15 min	Independent practice
5 min	Exit ticket

MINI-LESSON · TEACH FROM THE TEST-DAY SHEET

Three weeks, one reference sheet

WEEK 15

Area, volume, angles, and similar triangles

WEEK 16

Right triangles and trigonometry

WEEK 17

Circles and radians

On the official reference sheet

- Area of a circle, rectangle, and triangle
- Circumference of a circle
- The Pythagorean theorem
- 30-60-90 and 45-45-90 triangles
- Volumes of a box, cylinder, sphere, cone, and pyramid
- A circle has 360°, or 2π radians
- A triangle's angles sum to 180°

MINI-LESSON · NOT ON THE SHEET

What students must memorize

TOPIC	FORMULA OR FACT
Trig ratios	sin = opposite/hypotenuse, cos = adjacent/hypotenuse, tan = opposite/adjacent
Complementary angles	$\sin x° = \cos(90 - x)°$
Circle equation	$(x - h)^2 + (y - k)^2 = r^2$, center (h, k), radius r
Arc length	(central angle / 360°) × $2\pi r$
Sector area	(central angle / 360°) × πr^2
Degrees to radians	multiply by π/180
Similar figures	lengths × k, areas × k^2, volumes × k^3
Parallel lines and a transversal	alternate interior angles are equal; same-side interior angles sum to 180°

sin A and cos B

PROBLEM

In right triangle ABC, angle C is 90°, AC = 5, BC = 12, and AB = 13.

What is sin A? What is cos B?

SOLUTION

The side **opposite A** is BC = 12.

sin A = 12/13

The side **adjacent to B** is also BC = 12.

cos B = 12/13

They're equal because A and B are complementary: sin x° = cos(90 − x)°.

WORKED EXAMPLE 2 · CIRCLE EQUATION · WEEK 17

Center and radius

PROBLEM

Find the center and radius of:

$$x^2 + y^2 - 6x + 4y - 12 = 0$$

SOLUTION · COMPLETE THE SQUARE

Move the constant, then add (½ × coefficient)² to both sides:

$$(x^2 - 6x + 9) + (y^2 + 4y + 4) = 12 + 9 + 4$$

$$(x - 3)^2 + (y + 2)^2 = 25$$

Center (3, −2), radius 5

Desmos check: type the original equation and click the circle's center and edges.

A 60° central angle

PROBLEM

A circle has radius 6. A central angle measures 60°. Find the arc length and the angle in radians.

SOLUTION

ARC LENGTH

$$(60/360) \times 2\pi(6)$$

$$= 2\pi$$

RADIANS

$$60 \times \pi/180 = \pi/3$$

WORKED EXAMPLE 4 · SIMILAR SOLIDS · WEEK 15

Two similar cylinders

PROBLEM

Two similar cylinders have radii 2 and 6. The smaller has a volume of 10π cubic units. What is the larger cylinder's volume?

SOLUTION

Scale factor: 6 ÷ 2, so **k = 3**.

Volume scales by $k^3 = 3^3 = \mathbf{27}$.

10π × 27

= 270π cubic units

PRACTICE

Guided, then independent

GUIDED · 15 MIN

Multi-step problems on whiteboards

Before solving, students:

- Mark the diagram with every length and angle they can deduce.
- Say whether the needed formula is on the reference sheet.

INDEPENDENT · 15 MIN

Timed set each week

8–10 problems matching that week's focus

+2 review problems from earlier weeks

COMMON TRAPS

Where points get lost

01 Using the wrong side for opposite versus adjacent.

02 Not moving the constant before completing the square, or not adding to both sides.

03 Radius versus diameter in area, circumference, and the circle equation ($r^2 = 25$ means $r = 5$).

04 Scaling area by k instead of k^2 for similar figures.

05 Assuming a diagram is drawn to scale.

WRAP-UP · 5 MIN

Exit tickets by week

WEEK 15

One similar-figures problem

WEEK 16

One trig ratio and one complementary-angle problem

WEEK 17

One circle-equation problem

Source material

- College Board Question Bank: Geometry and Trigonometry
- Khan Academy explainers for the unit circle and right-triangle trig
- The official reference sheet from the College Board site

THE 1600 PROJECT · INTEGRATION THREAD

Unit 7

Mixed Problem Sets and Calculator Strategy

WEEKS	SAT DOMAINS	SESSION
18 to 20 and beyond	All four, mixed	60 minutes

AT A GLANCE

Goals for this unit

Students will be able to:

- Complete a 22-question math module in 35 minutes, the digital SAT's real timing.
- Decide in a few seconds whether to use Desmos or solve by hand.
- Use five core Desmos moves fluently.
- Track their own error patterns across all four domains.

ONE SAT MATH MODULE

22
questions

35
minutes

≈95 s
per question

CALCULATOR STRATEGY · 15 MIN · FIRST WEEK

Desmos or by hand?

Use Desmos when...

- Solving a system: graph both lines, click the intersection
- Finding roots or the vertex of a quadratic
- Checking whether two expressions are equivalent: graph both and see if they overlap
- An equation is hard to rearrange: graph each side, find where they meet
- Finding a regression or line of best fit from a table

Solve by hand when...

- It's a one-step linear equation
- It's a percent or ratio you can do mentally
- It's a definition or concept question (no computation)
- Typing the expression would take longer than the algebra
- Exact answers with radicals are needed and Desmos shows decimals (then confirm by hand)

CALCULATOR STRATEGY · DRILL THESE

Five Desmos moves

1

Intersection

Type both equations and click where they cross.

2

Zeros and vertex

Graph a quadratic and click the gray points.

3

Sliders

Type an equation with a constant like k, add a slider, and adjust it to see when a system has no solution or a quadratic has one root.

4

Equivalence check

Graph the original expression and each answer choice; the overlapping graph is the match.

5

Tables and regression

Enter data in a table, then type $y_1 \sim mx_1 + b$ for a line of best fit.

EVERY WEEK

The 60-minute mixed session

5 min

Warm-up

“Desmos or by hand?” Flash 6 problems; students vote and justify.

35 min

Timed module

22 mixed questions across all four domains, calculator allowed, in rough order of difficulty.

12 min

Error log

For every miss, record domain, topic, error type (concept, careless, time, calculator), and the fix.

8 min

Mini-reteach

The mentor re-teaches the most-missed topic across the group.

TEACH THIS ROUTINE

Pacing: two passes, no blanks

≈95 s

per question, on average

FIRST PASS	Answer everything you can in under 90 seconds; flag the rest.
SECOND PASS	Return to flagged items, trying Desmos first if you're stuck.
NO BLANKS	Never leave a blank. There is no guessing penalty.
FILL-INS	Check the answer format: fractions and decimals are fine, but no mixed numbers.

THE 1600 PROJECT · DESMOS TOOLKIT

Desmos for Algebra

SKILL AREAS	FOR EACH ONE	PAIRS WITH
5	A technique slide + a blank slide for yours	Lesson 1.1, Unit 7

HOW WE USE DESMOS

A tool, not the solver

1

Predict

Set up the problem and estimate the answer before touching the calculator.

2

Type

Enter exactly what's given. Let Desmos do the arithmetic and the graphing.

3

Check

Does the result match your prediction, and does it answer the question asked?

THREE RULES FOR EVERY PROBLEM

Rename the variables to **x** and **y**

= graphs; **~** fits a model to data

Type **_** for subscripts: x_1 gives x_1

BEFORE THE TECHNIQUES

Six Desmos essentials

Points of interest

Click a curve to show gray points at intercepts, maximums, minimums and intersections. Click a point to see its coordinates.

Hide and show

Click the colored circle next to an expression to turn its graph off or on.

Sliders

Type a letter Desmos doesn't know yet, like k, and it offers a slider. Drag it to watch the graph change.

Tables

Click + and choose “table.” The columns are named x_1 and y_1, and the points plot automatically.

Regression

Use ~ instead of = to fit a model to data, like y_1 ~ mx_1 + b. Desmos reports the constants.

Lists

Square brackets hold several values. f([1, 2, 3]) evaluates all of them at once.

TECHNIQUE · LINEAR EQUATIONS IN ONE VARIABLE

Graph both sides

Use it when: a one-variable equation is messy to rearrange (fractions, parentheses, x on both sides).

STEPS

1. Type the left side as y = (left side).
2. Type the right side as y = (right side).
3. Click the intersection. Its x-coordinate is the solution.
4. No intersection means no solution. One line on top of the other means infinitely many.

TRY IT

Solve $3(x - 2) + 5 = 2x + 9$.

```
y = 3(x-2) + 5
y = 2x + 9
```

→ Intersection (10, 29), so x = 10

WATCH OUT

Desmos graphs in x and y. If the equation uses another letter, like t or m, retype it with x.

TECHNIQUE · LINEAR FUNCTIONS AND EQUATIONS IN TWO VARIABLES

Let regression find m and b

Use it when: you have a table, two points, or values like f(2) = 7, and need the line.

STEPS

1. Click + and add a table. Enter the points in x_1 and y_1.
2. Type y_1 ~ mx_1 + b (tilde, not equals).
3. Read m and b from the regression results.
4. Define f(x) with those numbers, then type f(10) or any input to evaluate.

TRY IT

f(2) = 7 and f(5) = 16. What is f(10)?

```
table: (2, 7), (5, 16)
y_1 ~ mx_1 + b
f(x) = 3x + 1
f(10)
```

→ m = 3, b = 1, f(10) = 31

WATCH OUT

$R^2 = 1$ means the points are exactly linear. Anything less means a typo, or data that isn't linear.

TECHNIQUE · SYSTEMS OF TWO LINEAR EQUATIONS

Click the intersection

Use it when: the question gives two linear equations and asks for x, y, or a combination like x + y.

STEPS

1. Rename the variables to x and y, and write down which is which.
2. Type both equations exactly as given; no need to solve for y first.
3. Click the intersection point.
4. Answer the quantity the question asks for, not just the point.

TRY IT

a + s = 150 and 12a + 8s = 1,480. Find a.

```
x + y = 150
12x + 8y = 1480
```

→ (70, 80), so a = 70

WATCH OUT

Parallel lines: no solution. Same line: infinitely many. For “find k,” a slider shows the idea; confirm by matching slopes.

TECHNIQUE · LINEAR INEQUALITIES

Let the shading answer it

Use it when: you need the point that satisfies an inequality or a system of them.

STEPS

1. Type each inequality as given; Desmos shades its solutions.
2. For a system, the overlap of the shaded regions is the solution set.
3. Type each answer-choice point, like (2, 5), and see which one lands in the overlap.
4. One variable: type it with x only; the shaded strip is the answer.

TRY IT

Which point satisfies $y \le 2x + 3$ and $y > -x + 1$?

```
y ⩽ 2x + 3
y > -x + 1
(2, 5)
```

→ (2, 5) sits in the overlap

WATCH OUT

Dashed boundary = < or > (points on it don't count). Solid = ≤ or ≥.

TECHNIQUE · TESTING ANSWER CHOICES

Check all four at once

Use it when: the question gives a function or equation and four candidate values.

STEPS

1. Define the function, like f(x) = 4x − 7.
2. Type f( ) with the answer choices inside square brackets.
3. Desmos returns a list of outputs in the same order as the choices.
4. Pick the choice that gives the required value.

TRY IT

f(x) = 4x − 7. Which input gives an output of 13?

```
f(x) = 4x - 7
f([2, 3, 5, 8])
```

→ [1, 5, 13, 25], so x = 5

WATCH OUT

Use square brackets for lists and a one-letter function name like f or g. Works in every module, not only Algebra.

THE 1600 PROJECT · DESMOS TOOLKIT

Desmos for Advanced Math

SKILL AREAS
7

FOR EACH ONE
A technique slide + a blank slide for yours

PAIRS WITH
Lessons 2.1, 2.2, Unit 7

HOW WE USE DESMOS

A tool, not the solver

1

Predict

Set up the problem and estimate the answer before touching the calculator.

2

Type

Enter exactly what's given. Let Desmos do the arithmetic and the graphing.

3

Check

Does the result match your prediction, and does it answer the question asked?

THREE RULES FOR EVERY PROBLEM

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Type **_** for subscripts: x_1 gives x_1

BEFORE THE TECHNIQUES

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Hide and show

Click the colored circle next to an expression to turn its graph off or on.

Sliders

Type a letter Desmos doesn't know yet, like k, and it offers a slider. Drag it to watch the graph change.

Tables

Click + and choose “table.” The columns are named x_1 and y_1, and the points plot automatically.

Regression

Use ~ instead of = to fit a model to data, like y_1 ~ mx_1 + b. Desmos reports the constants.

Lists

Square brackets hold several values. f([1, 2, 3]) evaluates all of them at once.

TECHNIQUE · EQUIVALENT EXPRESSIONS

Overlap means equivalent

Use it when: the question asks which expression is equivalent to another.

STEPS

1. Type the original as y = (expression).
2. Type each answer choice as its own y = ...
3. Turn choices off one at a time (click the colored circle) until one sits exactly on the original.
4. Zoom out to confirm they match everywhere, not just near one point.

TRY IT

Which is equivalent to $(x + 3)^2 - 2(x + 3)$?

```
y = (x+3)^2 - 2(x+3)
y = x^2 + 4x + 3
y = x^2 + 4x + 9
```

→ $x^2 + 4x + 3$ overlaps exactly

WATCH OUT

Desmos doesn't mark holes: $(x^2 - 9)/(x - 3)$ overlaps $x + 3$ everywhere except $x = 3$.

TECHNIQUE · NONLINEAR EQUATIONS IN ONE VARIABLE

Count the crossings

Use it when: a quadratic, radical, rational or absolute-value equation, or “how many solutions?”

STEPS

1. Graph each side as y = ...
2. Click every intersection. The x-values are the solutions.
3. Number of intersections = number of real solutions.
4. Zoom out to make sure none are hiding off-screen.

TRY IT

Solve √(x + 7) = x − 5.

```
y = sqrt(x+7)
y = x - 5
```

→ One intersection, (9, 4), so x = 9

WATCH OUT

Desmos only graphs where an expression is defined, so extraneous roots (like x = 2 here) never show up.

TECHNIQUE · SYSTEMS WITH A NONLINEAR EQUATION

Read the points of intersection

Use it when: a line meets a parabola or circle, or you need a system's number of solutions.

STEPS

1. Type both equations as given.
2. Click each intersection point.
3. Two crossings = 2 solutions; a touch = 1; no contact = 0.
4. For “find k so there's exactly one solution,” put k on a slider and slide until the line just touches.

TRY IT

How many solutions: $y = x^2 - 4x + 1$ and $y = 2x - 8$?

```
y = x^2 - 4x + 1
y = 2x - 8
```

→ The line touches once at (3, −2): one solution

WATCH OUT

A slider gets you close, not exact. Confirm k by setting the discriminant equal to 0.

TECHNIQUE · QUADRATICS

Click the gray points

Use it when: you need zeros, the vertex, a max or min, or the y-intercept.

STEPS

1. Type the quadratic as y = ...
2. Click the curve. Gray points appear at the zeros, the vertex and the y-intercept.
3. Click each gray point to read its coordinates.
4. For the number of solutions, count x-intercepts, or replace c with a slider to explore.

TRY IT

Find the zeros and minimum of $y = x^2 - 5x - 14$.

```
y = x^2 - 5x - 14
```

→ Zeros (−2, 0) and (7, 0); vertex (2.5, −20.25)

WATCH OUT

The vertex's y-value is the min or max value; its x-value is where it happens. Answer the one asked.

TECHNIQUE · FUNCTION NOTATION AND TRANSFORMATIONS

Define it once

Use it when: f(x) notation, composition, or “how does the graph of g relate to f?”

STEPS

1. Define the function: f(x) = ...
2. Evaluate by typing f(5), f(g(2)) and so on.
3. To solve f(x) = 7, graph y = 7 and click its intersections with f.
4. Graph g(x) = f(x + 2) − 1 next to f to see the shift.

TRY IT

f(x) = (x − 3)² + 4 and g(x) = f(x + 2) − 1.

```
f(x) = (x-3)^2 + 4
g(x) = f(x+2) - 1
f(5)
```

→ f(5) = 8; g's vertex is (1, 3)

WATCH OUT

f(2) means evaluate; f(x) = 2 means solve. Type f(2) for the first, graph y = 2 for the second.

TECHNIQUE · EXPONENTIAL MODELS

Type the model, read the value

Use it when: a growth or decay word problem, or “which model fits this table?”

STEPS

1. Type the model as a function, like V(x) = 24000(0.85)^x.
2. Evaluate it directly: V(2).
3. To find when it reaches a value, graph y = (target) and click the intersection.
4. For a table, enter the data and type y_1 ~ ab^(x_1) to get a and b.

TRY IT

A $24,000 car loses 15% per year. Value after 2 years?

```
V(x) = 24000(0.85)^x
V(2)
```

→ 17340, so $17,340

WATCH OUT

In a regression, b > 1 means growth and b < 1 means decay. A 15% decrease is 0.85, not 0.15.

TECHNIQUE · FINDING UNKNOWN CONSTANTS

Let regression find a and b

Use it when: an identity like $a(x + 3)^2 + b = 2x^2 + 12x + 25$ holds “for all x.”

STEPS

1. Make a list of x-values: x_1 = [0, 1, 2, 3, 4].
2. Make y-values from the known side: y_1 = 2x_1^2 + 12x_1 + 25.
3. Regress with the unknown side: y_1 ~ a(x_1 + 3)^2 + b.
4. Read a and b. $R^2 = 1$ confirms it's an exact identity.

TRY IT

$a(x + 3)^2 + b = 2x^2 + 12x + 25$ for all x. Find b.

```
x_1 = [0,1,2,3,4]
y_1 = 2x_1^2 + 12x_1 + 25
y_1 ~ a(x_1+3)^2 + b
```

→ a = 2, b = 7

WATCH OUT

Use more x-values than unknowns. If R^2 isn't exactly 1, check your typing.